GRADUATION PROJECT REPORT

MediNova-AI

A Web-Based Healthcare Assistance System

GitHub Repository:
https://github.com/Alaa-AHK/MediNova-AI

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1. Project Planning & Management

1.1 Project Proposal

MediNova-AI is a web-based healthcare assistance platform designed to provide users with access to multiple health-related services through a unified and user-friendly interface. The system enables users to select a service of interest and receive relevant outputs or predictions based on the selected service area.

The project is divided into four main service modules:

- Radiology: Assists in analyzing radiological images such as X-rays.
- Pathology: Provides support related to pathology data and reports.
- Prescription: Offers assistance in interpreting or generating basic prescription information.
- Skin Diseases: Supports the identification of common skin conditions based on user input or images.

Each service module is developed independently by a dedicated sub-team. The system is intended to serve as a supportive tool that complements the work of medical professionals, and is not designed to replace clinical diagnosis or medical advice. The project emphasizes clean user interface design, reliable service selection, and structured output presentation.

1.2 Objectives

The primary objectives of the MediNova-AI project are as follows:

1. To develop a web-based platform that allows users to access multiple healthcare-related services from a single interface.
2. To implement four separate service modules (Radiology, Pathology, Prescription, and Skin Diseases), each developed independently.
3. To design an intuitive and accessible user interface that guides users through service selection and result interpretation.
4. To ensure each service module delivers relevant outputs based on user inputs.
5. To demonstrate the potential of technology-assisted healthcare support in an academic setting.
6. To provide a foundation that can be extended or improved in future development cycles.

1.3 Scope

The scope of this project covers the design, development, and basic testing of a web-based healthcare assistance system. The following boundaries define what is and is not included within the scope of this project:

In Scope:

- A functional web interface allowing users to select from four healthcare service modules.
- Individual module pages for Radiology, Pathology, Prescription, and Skin Diseases.
- Basic input handling and output display for each module.
- Unit testing and basic usability testing.

Out of Scope:

- Clinical validation or medical certification.
- Mobile application development.
- Multi-language support or accessibility compliance beyond basic usability.

It is important to emphasize that MediNova-AI is a supportive system and is not intended to replace qualified medical professionals. All outputs generated by the system should be treated as informational only and must not be used as a substitute for professional medical consultation.

1.4 Project Timeline

The following table outlines the planned project timeline from initiation to final deployment. The timeline is designed to be realistic, allowing adequate time for each phase of development.

Phase	Key Activities	Duration	Responsible
Research Phase	Review literature, study existing healthcare systems, identify project requirements and constraints.	2 weeks	All Team Members
Data Collection / Dataset Selection	Identify and collect relevant datasets or sample data for each service module. Evaluate data quality and suitability.	2 weeks	Module Sub-Teams
Algorithm Selection	Evaluate and select appropriate processing methods or algorithms for each module based on the data type and output requirements.	1 week	Module Sub-Teams
System Design	Define system architecture, module structure, database schema (if applicable), and interface flow diagrams.	1 week	All Team Members
Implementation Phase	Develop the frontend and service modules. Each sub-team implements their assigned module independently.	4 weeks	Module Sub-Teams
Testing Phase	Conduct unit testing per module, integration testing of service selection flow, and usability testing with sample users.	2 weeks	All Team Members
Finalization & Deployment	Fix identified bugs, finalize documentation, prepare presentation, and deploy the system to a local or basic hosting environment.	1 week	All Team Members

Total estimated project duration: approximately 14 weeks.

1.5 Task Assignment

The project team is divided into four sub-groups, each responsible for one of the four service modules. This structure ensures clear ownership, parallel development, and focused expertise for each service area.

Group	Service Module	Primary Responsibilities	Shared Responsibilities
Group 1	Radiology	Radiology module UI, input handling, output display for radiological data.	System integration, documentation, testing.

Group	Service Module	Primary Responsibilities	Shared Responsibilities
Group 2	Pathology	Pathology module UI, input handling, output display for pathology data.	System integration, documentation, testing.
Group 3	Prescription	Prescription module UI, input handling, output display for prescription assistance.	System integration, documentation, testing.
Group 4	Skin Diseases	Skin diseases module UI, input handling, output display for skin condition support.	System integration, documentation, testing.

In addition to module-specific development, all team members contribute collectively to the overall system design, final integration, project documentation, and presentation preparation.

1.6 Risk Management

The following table identifies potential risks associated with the project, their likelihood, potential impact, and proposed mitigation strategies.

Risk	Likelihood	Impact	Mitigation Strategy
Inconsistent development progress across sub-teams	Medium	High	Weekly progress check-ins; shared project board to track tasks and milestones.
Insufficient or low-quality dataset for a module	Medium	Medium	Identify multiple dataset sources early. Use publicly available datasets as fallback options.
Integration issues between modules and the main interface	Low	High	Define a common interface contract early. Conduct integration testing after each module is completed.
Scope creep	Medium	Medium	Strictly adhere to the defined scope. Any additions must be approved by the full team and supervisor.
Team member availability issues	Low	Medium	Cross-train team members on adjacent modules to provide backup support when needed.
Technical difficulties with hosting or deployment	Low	Low	Use local hosting as primary fallback. Test deployment environment well before the submission deadline.

2. Literature Review

2.1 Overview of AI in Healthcare

The application of artificial intelligence in the healthcare sector has grown considerably over the past decade. AI-based tools have been explored for a wide range of medical tasks, including diagnostic imaging analysis, clinical decision support, drug discovery, and patient monitoring. Research institutions and technology companies alike have invested in developing systems that can assist clinicians in processing large volumes of medical data more efficiently.

Despite significant progress, the majority of AI healthcare applications remain assistive in nature. They are designed to support, not replace, the clinical judgment of medical professionals. This distinction is important when framing the design and purpose of MediNova-AI, which adopts the same supportive philosophy.

2.2 Radiology Systems

Radiology has been one of the most active areas for AI-assisted tools in healthcare. Automated analysis of chest X-rays, CT scans, and MRI images has been studied extensively, with several published models demonstrating the ability to detect abnormalities such as pneumonia, tumors, and fractures.

Key challenges in this domain include variation in image quality, the need for large annotated training datasets, and the difficulty of generalizing models across different imaging equipment. Several open-source datasets, such as the NIH Chest X-ray Dataset, have been made available to support research in this area. The MediNova-AI radiology module draws on this body of work as a reference for its design and expected functionality.

2.3 Skin Disease Detection Systems

Dermatology is another area where AI-assisted tools have shown promising results. Studies have demonstrated that classification models trained on labeled skin lesion datasets can achieve accuracy levels comparable to those of general practitioners for certain conditions. Projects such as the ISIC (International Skin Imaging Collaboration) challenge have contributed standardized datasets and benchmarks for evaluating such systems.

The MediNova-AI skin diseases module is informed by this research landscape. The module provides a simplified interface for users to receive informational outputs related to common skin conditions, without claiming clinical-grade diagnostic accuracy.

2.4 Pathology-Related Systems

Computational pathology involves the use of digital image analysis and data processing techniques to assist in the interpretation of pathological specimens. Research in this area has explored the automation of cell counting, tissue classification, and abnormality detection in histopathological slides.

While highly specialized tools exist in research and some clinical environments, access to such systems remains limited in many settings. The MediNova-AI pathology module provides a simplified representation of this service area, focusing on user interaction and structured output rather than advanced image processing.

2.5 Prescription Assistance Systems

Prescription assistance tools have been developed to help both patients and clinicians better understand medication-related information. These systems may include drug interaction checkers, dosage calculators, and medication lookup tools. Several commercial and open-access platforms provide such functionalities.

The MediNova-AI prescription module focuses on providing a structured interface for users to access prescription-related information. The module is informational and does not generate actual medical prescriptions or replace the role of a licensed pharmacist or physician.

2.6 Challenges

Several challenges are commonly identified across AI-assisted healthcare research and development:

- Data Availability and Quality: Obtaining sufficient, well-labeled medical data for training and evaluation is often difficult due to privacy regulations and the cost of expert annotation.
- Accuracy and Reliability: No automated system achieves perfect accuracy. Errors in a medical context can have serious consequences, which demands rigorous validation before deployment in real clinical settings.
- Ethical and Legal Considerations: Questions around patient consent, data privacy, liability in case of error, and the transparency of algorithmic decisions remain active areas of debate.
- User Adoption: Healthcare professionals may be cautious about adopting AI tools, particularly when the reasoning behind system outputs is not transparent or explainable.
- Generalizability: Models trained on data from one population or setting may not perform equally well when applied to different demographics or healthcare environments.

The MediNova-AI project acknowledges these challenges. By limiting the scope to a supportive and educational interface, the project avoids overstating its capabilities and remains grounded in realistic expectations.

3. Requirements Gathering

3.1 Stakeholders

The following stakeholders have been identified for the MediNova-AI project:

Stakeholder	Role and Interest
End Users (General Public)	Individuals who access the system to obtain health-related information and service outputs. They expect a clear, accessible, and easy-to-use interface.
Project Team	The four sub-teams responsible for designing, developing, and testing the system. Each sub-team owns a specific service module.
Instructor	Provides guidance on project direction, evaluates deliverables, and ensures the project meets academic standards.
Healthcare Professionals (Reference)	Not direct users of the system, but referenced as domain experts whose workflows and needs influenced the design of each service module.

3.2 User Stories

The following user stories describe the expected interactions between users and the MediNova-AI system. These stories are realistic and aligned with the current scope of the project.

General User Stories:

- As a user, I want to access a home page that clearly explains the available services, so that I can quickly understand what the system offers.
- As a user, I want to select a specific healthcare service from the main menu, so that I can navigate to the relevant module.
- As a user, I want to submit input relevant to the chosen service (such as uploading an image or filling out a form), so that I can receive an appropriate output.
- As a user, I want to receive a clear and readable output from the selected service module, so that I can understand the result presented.
- As a user, I want the system to load quickly and respond without long delays, so that I can complete my interaction efficiently.

Module-Specific User Stories:

- Radiology: As a user, I want to upload a radiological image and receive a structured output indicating relevant findings or observations.
- Pathology: As a user, I want to enter or upload pathology-related data and receive an organized summary or result.
- Prescription: As a user, I want to input prescription-related details and receive relevant informational output to help me better understand medication information.
- Skin Diseases: As a user, I want to upload or describe a skin condition and receive informational output about potential conditions.

3.3 Functional Requirements

The following functional requirements define what the MediNova-AI system must be capable of doing. These requirements are based on the current project scope, which focuses on service selection.

FR-1: Home Page

- The system shall display a home page that introduces MediNova-AI and lists the four available service modules.
- The home page shall include navigation options to access each service module.

FR-2: Service Selection

- The system shall allow users to select one of the four service modules: Radiology, Pathology, Prescription, or Skin Diseases.
- Each service module shall be accessible from the main navigation interface.

FR-3: Radiology Module

- The radiology module shall provide an interface for users to upload or submit radiological input.
- The module shall display a structured output in response to the submitted input.

FR-4: Pathology Module

- The pathology module shall provide an interface for users to input or upload pathology-related data.
- The module shall display relevant output based on the submitted information.

FR-5: Prescription Module

- The prescription module shall provide a form-based interface for entering prescription-related details.
- The module shall display informational output relevant to the entered data.

FR-6: Skin Diseases Module

- The skin diseases module shall provide an interface for uploading or describing a skin condition.
- The module shall return an informational response based on the user submission.

FR-7: Output Display

- All modules shall present outputs in a clear, structured, and readable format.
- The system shall not display raw or unformatted data to the user.

3.4 Non-Functional Requirements

Category	Requirement ID	Description
Performance	NFR-01	The system shall load any service module page within 3 seconds under normal network conditions.
Usability	NFR-02	The interface shall be intuitive and navigable by users without technical training.
Reliability	NFR-03	The system shall remain operational throughout a user session without unexpected crashes or errors.
Maintainability	NFR-04	Each module shall be developed independently, allowing updates to one module without affecting others.
Compatibility	NFR-05	The system shall be accessible on standard desktop web browsers (Chrome, Firefox, Edge) without additional plugins.
Security	NFR-06	User-submitted data shall not be stored permanently or shared outside the session scope.
Scalability	NFR-07	The system architecture shall allow the addition of new service modules in future development phases without restructuring the entire system.

4. System Analysis & Design

4.1 Problem Statement

Access to preliminary health information and medical service support is often fragmented and difficult to navigate for general users. Individuals seeking information across multiple medical domains, such as radiology, pathology, prescriptions, and skin conditions, must typically consult different platforms or professionals for each concern.

MediNova-AI addresses this gap by providing a single, accessible web-based platform that consolidates four distinct healthcare service modules under one interface. The system aims to reduce the effort required to find relevant medical information and deliver structured, service-specific outputs to users in a clear and non-technical manner.

The problem is not a lack of medical information online, but rather the absence of a centralized, user-friendly system that organizes multiple healthcare services in one place and presents results in a structured, consistent format.

4.2 System Objectives

Based on the problem statement, the MediNova-AI system is designed to achieve the following objectives:

7. Provide a single entry point for users to access multiple healthcare-related service modules.
8. Deliver a consistent and intuitive user experience across all four service areas.
9. Enable each module to independently process user input and return structured output.

10. Maintain a clean separation between modules to facilitate independent development and future maintenance.
11. Present all outputs in a format that is readable and meaningful to non-technical users.

4.3 System Architecture

The MediNova-AI system follows a modular, frontend-focused architecture. The design prioritizes simplicity, maintainability, and independent module development. The architecture consists of two main layers:

Layer 1: Frontend (User Interface)

The frontend layer is the primary component of the system. It is responsible for rendering the interface, handling user interactions, routing between pages, and displaying outputs. The frontend is built using standard web technologies and includes the following components:

- Home Page: Introduces the system and provides navigation to the four service modules.
- Service Selection Interface: Presents the available modules in a clear and organized layout.
- Module Pages: Four individual pages, one for each service (Radiology, Pathology, Prescription, Skin Diseases), each containing its own input form and output display area.

Layer 2: Service Modules

Each service module operates as an independent unit. Modules handle their own input validation, processing logic, and output formatting. They do not share a common backend engine and are not connected to one another.

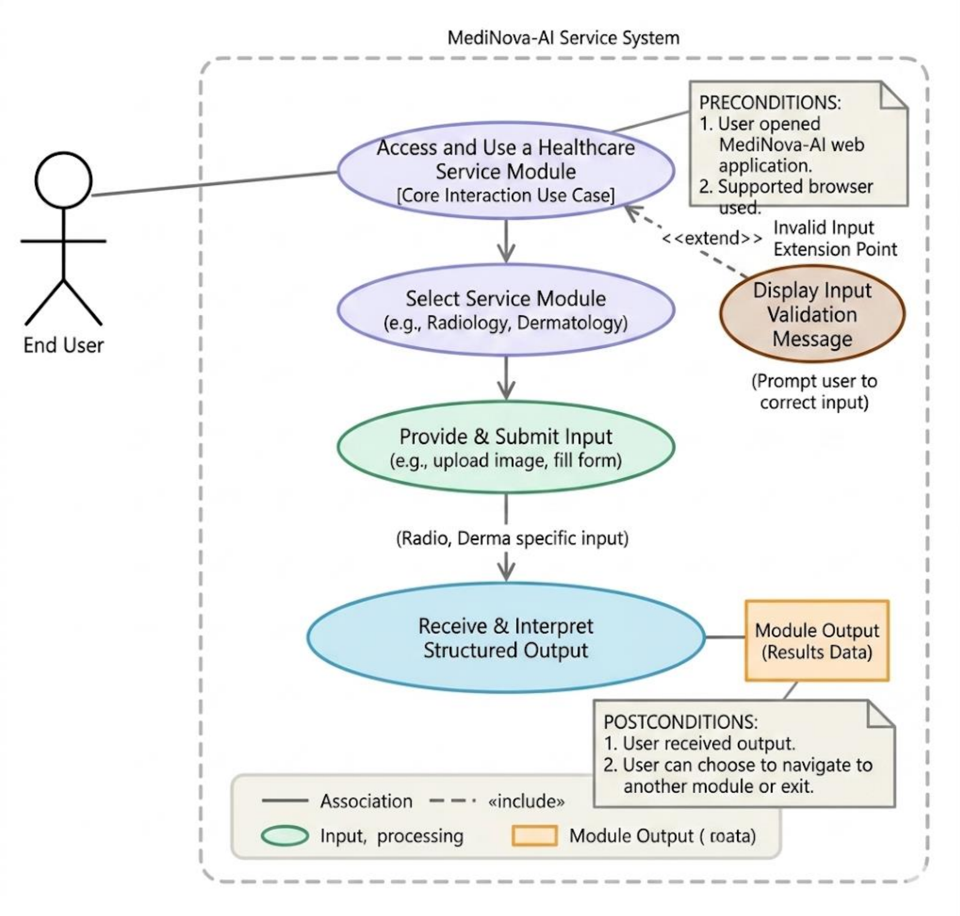
- Radiology Module: Processes radiological input and returns structured observations.
- Pathology Module: Handles pathology data input and provides relevant output.
- Prescription Module: Processes prescription-related form input and returns informational output.
- Skin Diseases Module: Handles skin condition input and returns informational response.

The architecture avoids a monolithic backend design. This approach reduces complexity, simplifies testing, and allows each sub-team to work on their assigned module independently without creating dependencies between modules.

4.4 Use Case Description

Field	Description
Use Case Name	Access and Use a Healthcare Service Module
Actor	End User
Precondition	The user has opened the MediNova-AI web application in a supported browser.
Main Flow	1. The user visits the home page and reviews the available services. 2. The user selects a service module (e.g., Radiology). 3. The system navigates the user to the selected module page. 4. The user enters the required input (e.g., uploads an image or fills a form). 5. The user submits the input. 6. The system processes the input and displays a structured output. 7. The user reads and interprets the output.
Alternate Flow	If the user submits invalid or incomplete input, the system displays a validation message prompting the user to correct the input before resubmitting.

Field	Description
Postcondition	The user has received output from the selected service module and may choose to navigate to another module or exit the system.



4.5 Data Flow Description (DFD)

The following describes the data flow within the MediNova-AI system at a conceptual level:

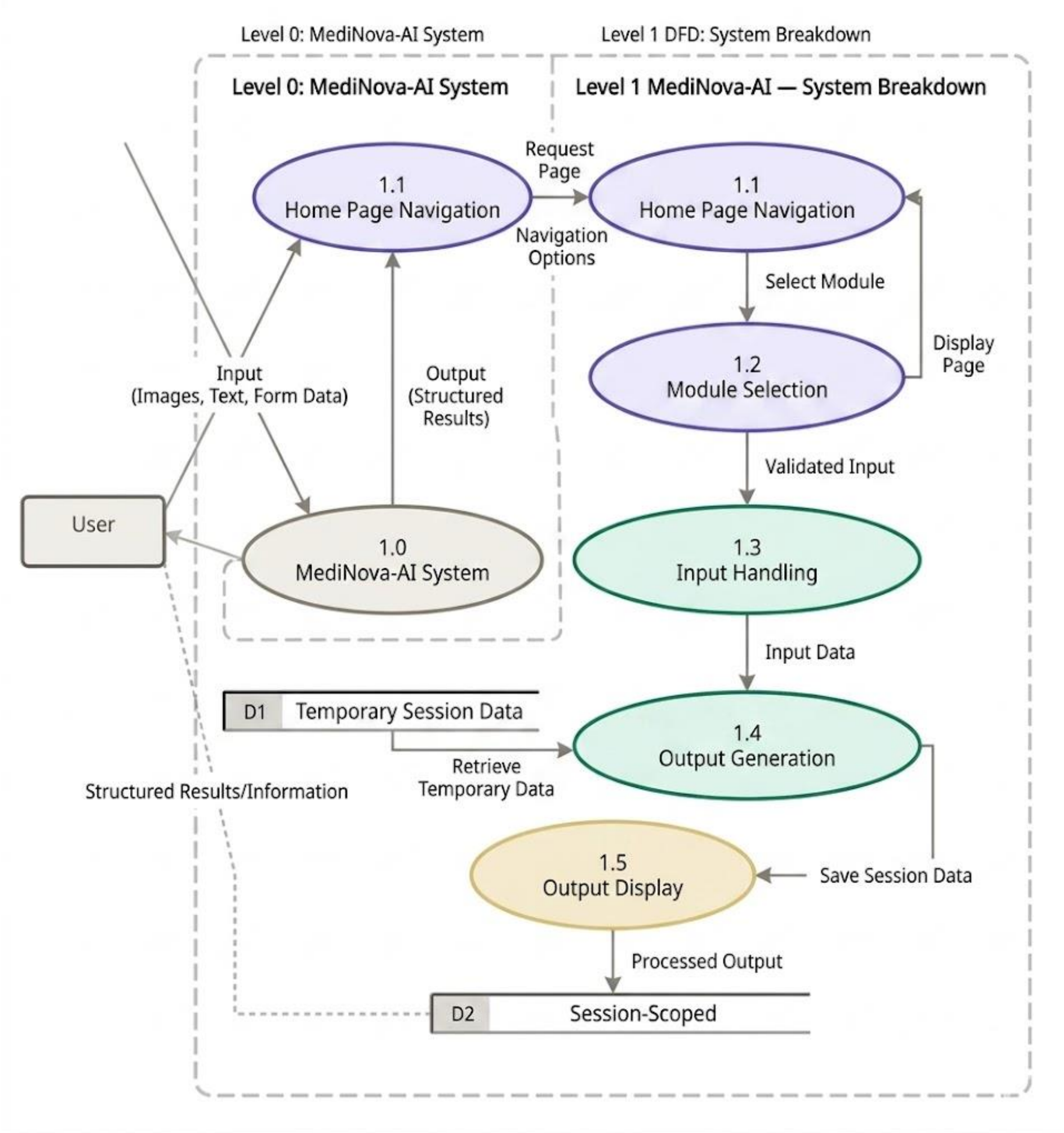
Level 0 (Context Diagram):

The user interacts with the MediNova-AI system by submitting input (images, text, or form data). The system processes this input within the selected service module and returns output (structured results or information) to the user. No external systems are involved.

Level 1 (Top-Level DFD):

- Process 1 - Home Page Navigation: The user accesses the home page. The system presents the list of available service modules. The user selects a module.
- Process 2 - Input Handling: The selected module receives user input. Input validation is performed. Valid input is passed to the processing component.
- Process 3 - Output Generation: The module processes the validated input and generates a structured output.
- Process 4 - Output Display: The system presents the generated output to the user on the module result page.

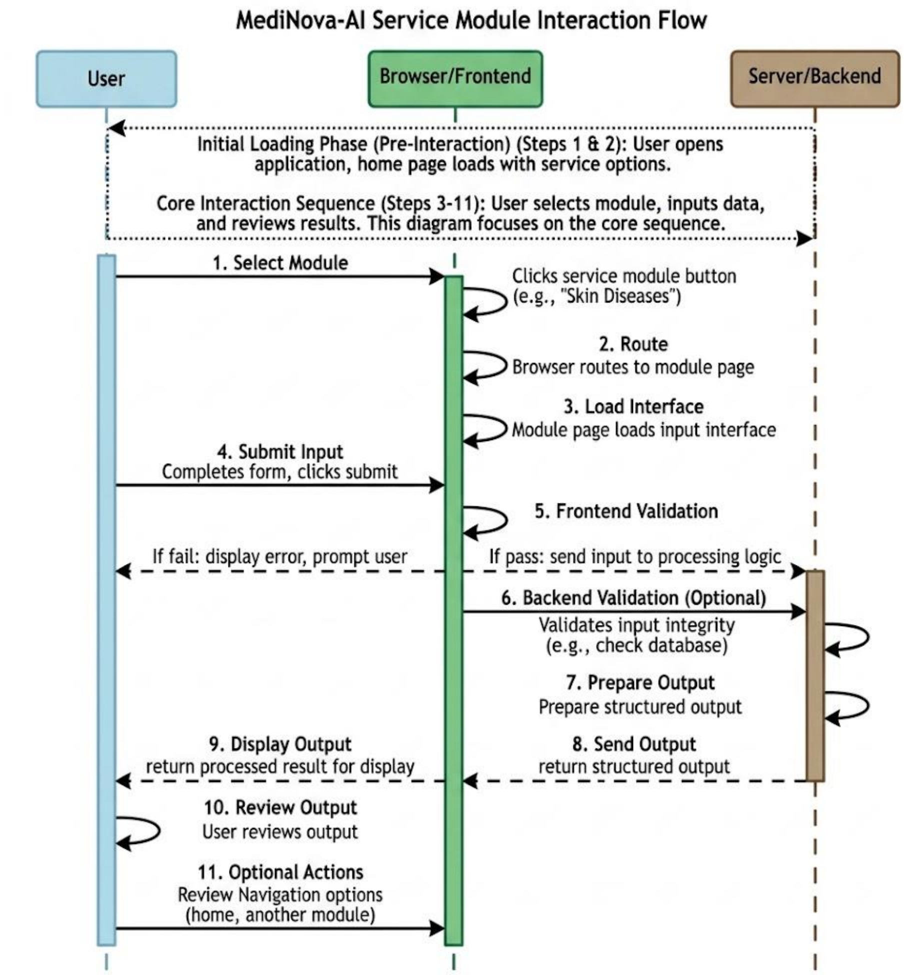
Data stores are minimal in this system. If temporary data storage is used, it is session-scoped and not persisted beyond the user's active session.



4.6 Sequence Flow

The following sequence describes the step-by-step interaction flow between the user and the system:

12. The user opens the MediNova-AI application in a web browser.
13. The home page loads and displays an overview of the system and available service modules.
14. The user clicks on a service module button (e.g., "Skin Diseases").
15. The browser routes the user to the Skin Diseases module page.
16. The module page loads its input interface (e.g., an image upload field or a description form).
17. The user completes the input form and clicks the submit button.
18. The frontend performs basic input validation. If validation fails, an error message is shown and the user is prompted to correct the input.
19. If validation passes, the input is sent to the module's processing logic.
20. The module processes the input and prepares a structured output.
21. The output is displayed to the user on the same page or a dedicated result section.
22. The user reviews the output. Navigation options allow the user to return to the home page or access another module.



4.7 Class Diagram Description

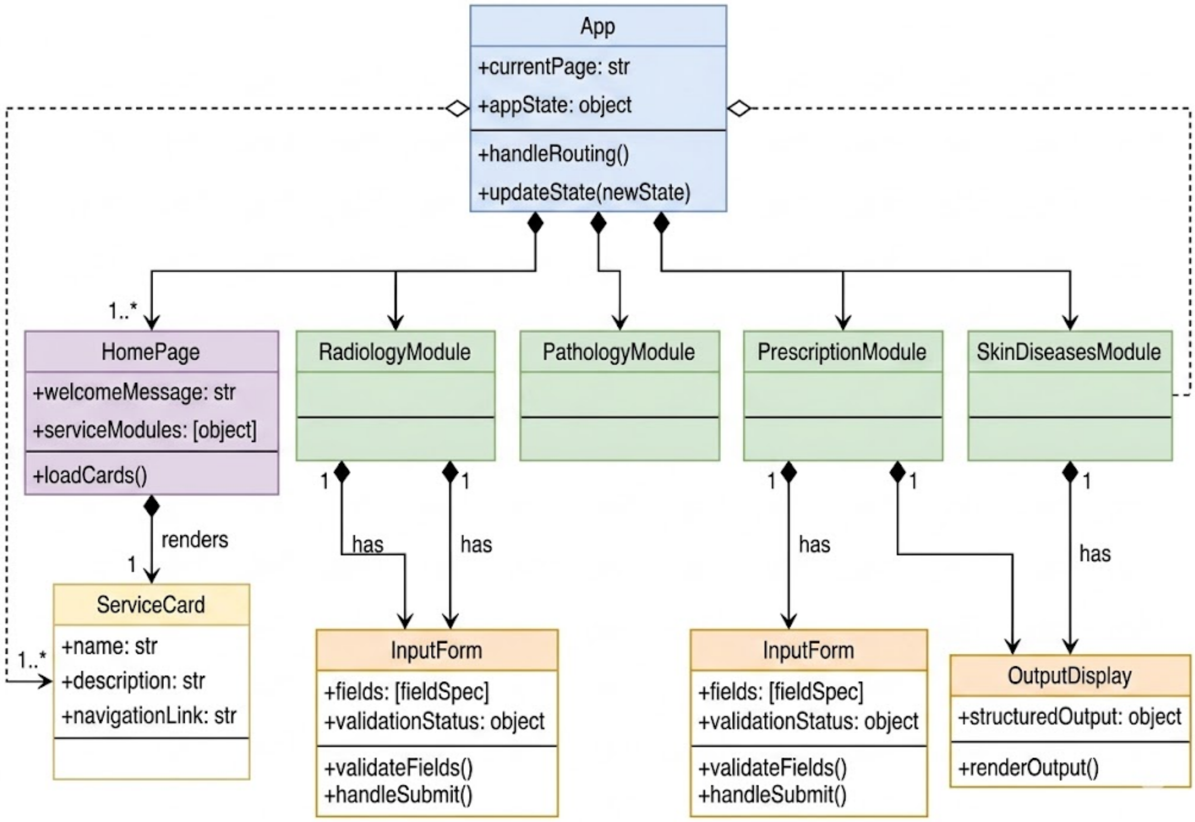
Given the frontend-focused nature of MediNova-AI, the class structure is relatively simple. The following describes the main logical components and their relationships:

Main Components:

- App: The root component of the application. Manages routing between pages and holds the overall application state.
- HomePage: Renders the home interface, including the welcome message and service selection cards. Has a dependency on ServiceCard.
- ServiceCard: A reusable UI component that represents a single service module option. Contains the module name, a brief description, and a navigation link.
- RadiologyModule: Handles the Radiology service page, including the input interface and output display. Contains InputForm and OutputDisplay as sub-components.
- PathologyModule: Handles the Pathology service page. Contains InputForm and OutputDisplay as sub-components.
- PrescriptionModule: Handles the Prescription service page. Contains InputForm and OutputDisplay as sub-components.
- SkinDiseasesModule: Handles the Skin Diseases service page. Contains InputForm and OutputDisplay as sub-components.
- InputForm: A reusable component that renders input fields appropriate for a given module. Handles validation and submission events.
- OutputDisplay: A reusable component that renders the structured output returned by a module.

Relationships:

- App contains and routes between HomePage and all four module pages.
- Each module page contains one InputForm and one OutputDisplay.
- HomePage contains multiple ServiceCard instances, one for each module.



4.8 Technology Stack

The MediNova-AI system is built using a frontend-focused technology stack. The technologies selected prioritize simplicity, broad compatibility, and ease of development.

Layer	Technology	Purpose
Frontend Structure	HTML5	Page structure and content markup for all module pages.
Frontend Styling	CSS3	Visual design, layout, and responsive styling across all pages.
Frontend Interactivity	JavaScript (Vanilla or lightweight framework)	Input handling, validation, dynamic output rendering, and page routing.
Version Control	Git / GitHub	Source code management, collaboration, and project versioning.
Development Environment	Visual Studio Code	Primary code editor used by all team members.
Testing	Browser Developer Tools	Manual testing, debugging, and performance inspection.

A lightweight backend component (such as a simple Node.js or Python server) may be used if module processing logic requires server-side execution. However, the core user-facing functionality is delivered through the frontend layer.

4.9 Deployment

The deployment strategy for MediNova-AI is intentionally simple and appropriate for an academic project. The following options are considered:

Primary Option - Local Deployment:

The system is deployed and demonstrated on a local machine. This approach ensures full control over the environment, eliminates network dependency during the demonstration, and requires no cost or external configuration. The application is accessed via a local web server (such as a simple HTTP server or a development server provided by the chosen framework).

Secondary Option - Basic Web Hosting:

If online accessibility is required, the system can be hosted on a basic free hosting platform such as GitHub Pages (for static content) or a low-cost shared hosting service. This option makes the system accessible from any browser without requiring local setup.

Deployment Steps:

23. Complete final testing and bug fixes.
24. Prepare a clean build of the project (remove development files and unused assets).
25. Configure the server or hosting environment.
26. Upload or deploy project files to the target environment.
27. Verify that all pages and modules are accessible and functional in the deployed environment.
28. Document the deployment URL or local setup instructions for submission.